FLIPKART GRIDLOCK HACKATHON 2.0 SOLUTION APPROACH DOCUMENT

Traffic Demand Prediction

1. PROBLEM STATEMENT

Predict traffic demand for geohash-encoded road segments at 15-minute intervals.
Target Variable: demand (continuous, range 0 to 1)
Evaluation Metric: score = max(0, 100 * R2_score(actual, predicted))
Training Data: Days 44 to 49 (multiple geohashes x 96 time intervals per day)
Test Data: Day 49 (remaining time intervals to predict)

2. OVERALL APPROACH: 2-LAYER STACKING ENSEMBLE

Our final solution uses a 2-layer stacking architecture with optimal blending:
Layer 1 (Base Models): 16 diverse tree-based regressors trained with 5-Fold Cross Validation.
Layer 2 (Meta-Learner): BayesianRidge regressor to optimally combine Out-Of-Fold predictions.
Final Output: Weighted blend of two stacks for variance reduction.

3. FEATURE ENGINEERING

Temporal Features

- Parsed timestamp into hour, minute, time_in_minutes, time_idx (15-min slot index).
- Cyclical sine/cosine encoding: hour_sin, hour_cos, minute_sin, minute_cos to capture the circular nature of time.
- Time period categorical features: late_night, early_morning, morning, afternoon, evening, night.
- Day of week features: day_of_week, day_of_week_sin, day_of_week_cos.

Spatial Features

- Decoded geohash to latitude and longitude using the geohash2 library.
- Spatial neighborhood aggregation at precision 4 and 5 (mean demand per geohash prefix and prefix+hour).
- Geohash prefix columns: geohash_4, geohash_5 as categorical features.

Historical Demand Features

- Per-geohash statistics: mean, std, median, max, min, count of demand.
- Per-geohash+hour statistics: mean and standard deviation.
- Day 48 exact demand at same geohash and timestamp. This direct previous-day lookup is the single most important feature.
- Time lags of 15 and 30 minutes from Day 48.
- Multi-day demand references: exact demand from days 44, 45, 46, and 47 at the same geohash and timestamp.
- Day-over-day ratios: d48/d47 ratio, d48/d46 ratio.
- Multi-day aggregates: mean, standard deviation, and trend.
- Sequential modeling features: last known demand, time gap from last known demand, multi-step lags, and rolling means.

Early Morning Calibration (Critical Innovation)

- Calculated early_morning_ratio = demand_day49_early / demand_day48_early for each geohash during the first 2 hours of the day.
- This captures the macroscopic day-level traffic shift between Day 48 and Day 49.
- Used this ratio to adjust the Day 48 exact demand.
- Also computed residuals, linear slope, and standard deviation to capture early-morning patterns.
- Ratio was clipped to prevent extreme extrapolation.

Demand Dynamics Features

- Demand velocity: rate of change at 15-min and 30-min resolution.
- Demand acceleration: second derivative (velocity of velocity).
- Demand range: max-min spread in a 30-min window.
- Demand momentum: deviation of previous-day demand from historical mean.

Road and Context Features

- Target encoding of RoadType and LargeVehicles.
- Interaction features: temperature x lanes, lanes squared, highway x hour.
- Binary indicators: is_highway, is_street, high_capacity.

Missing Value Handling

- Temperature: hierarchical imputation (geohash+hour, then hour, then global mean).
- RoadType and Weather: geohash-based lookup, filling remaining with Unknown.

4. MODEL ARCHITECTURE: 16-MODEL STACKING ENSEMBLE

Base Models (Layer 1) 5-Fold Cross-Validation

Models are designed for maximum diversity across algorithms, loss functions, target transforms, and hyperparameters:

1. CatBoost RMSE (primary workhorse)

2. LightGBM RMSE (leaf-wise)

3. XGBoost RMSE (depth-wise)

4. CatBoost Deep (different bias/variance)

5. LightGBM DART (dropout regularization)

6. ExtraTrees (zero gradient random forest variant)

7. CatBoost on log-transformed target

8. LightGBM on sqrt-transformed target

9. CatBoost MAE (L1 loss, robust to outliers)

10. LightGBM Huber (smooth L1/L2 hybrid)

11. XGBoost MAE

12. CatBoost Quantile (predicts median)

13. CatBoost Shallow (high bias, low variance)

14. LightGBM GOSS (Gradient-based One-Side Sampling)

15. HistGradientBoostingRegressor (fast, robust native sklearn tree)

16. CatBoost MAPE (Mean Absolute Percentage Error loss - tailored for demand scaling)

All models use early stopping on the validation set.

Meta-Learner (Layer 2)

- Stack A: BayesianRidge on predictions of original 8 baseline models.
- Stack C: BayesianRidge on predictions of all 16 models.
- BayesianRidge was chosen for automatic regularization via Bayesian priors to prevent overfitting to stacking features.

Final Blending

Final prediction = Weighted blend between Stack_A and Stack_C

Rationale: The 16-model stack captures a highly diverse range of decision boundaries. The baseline 8-model stack acts as a regularizer to stabilize the final ensemble predictions.

5. KEY INSIGHTS AND DESIGN DECISIONS

- The Day 48 lookup is the most predictive feature. The model learns that today's demand is mostly yesterday's demand, adjusted.
- Early morning calibration captures the day-level shift. Without this, models fail when Day 49 is globally busier or quieter than Day 48.
- Model diversity matters more than individual model strength. Adding models with MAE, Huber, Quantile, and MAPE loss functions consistently improved the ensemble.
- Blending a conservative stack with an aggressive stack is better than using either alone.
- Pseudo-labeling was tested but consistently hurt the score, introducing noise due to distribution shifts.

6. TOOLS AND ENVIRONMENT

Language: Python 3.10

Data Processing: Pandas, NumPy, Geohash2

Machine Learning: Scikit-Learn, LightGBM, CatBoost, XGBoost

Validation: 5-Fold KFold Cross-Validation (seed 42)

Environment: Google Colab (T4 GPU)

7. REPRODUCIBILITY

- All random seeds are fixed.
- Feature engineering is deterministic.
- Code is self-contained in a single Python script.